

Lecture Notes

Stochastic Hydrology

Module 3: Time Series Analysis & Geostatistics

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Contents

Preface	III
6 Time Series Analysis	1
6.1 Hydrologic Time Series	1
6.1.1 Stationarity	2
6.1.2 Ergodicity	2
6.2 Determination and Testing for Trends	3
6.2.1 Mann-Kendall Test for Trends - Nonparametric	3
6.2.2 Kendall's Correlation Coefficient	4
6.2.3 Linear Regression - Parametric	4
6.2.4 Kendall-Theil Linear Robust Line - Nonparametric	8
6.3 Determination and Testing for Shifts	8
6.4 Testing for Seasonality / Periodicity	9
6.5 Removal of Nonstationarities	9
6.5.1 Removal of Trends	9
6.5.2 Removal of Shifts	9
6.5.3 Removal of Periodicity	10
7 Stochastic Models	11
7.1 Autocorrelation	11
7.2 Steps in Setting up a Stochastic Model	12
7.3 Autoregressive AR Models	12
7.3.1 Autoregressive Model of Order 1 - AR(1)	13
7.3.2 Autoregressive Model of Order 2 - AR(2)	13
7.4 Moving Average MA Model	14
7.4.1 Moving Average Model of Order 1 - MA(1)	14
7.4.2 Moving Average Model of Order 2 - MA(2)	14
7.5 Autoregressive Moving Average ARMA Model	15
7.5.1 Autoregressive Moving Average Model of Order (1,1) - ARMA(1,1)	15
7.6 Partial Autocorrelation Function PACF	15
7.7 Verification of the Model Choice	17
7.7.1 Test for Independence of Residuals	18
7.7.2 Test for Normality of Residuals	18
7.8 Generation of Synthetic Series with ARMA Models	18
7.9 Multivariate Models	19

Bibliography	20
A Appendix	A1
A.1 Tables of Critical Values	A1

Preface

The content and structure of these lecture notes is partially based on the following works:

Handbook of Hydrology

McGraw-Hill, Inc. 1992. Maidment, David R., ed.

These lecture notes are intended for personal use in the Stochastic Hydrology course and may not be distributed further.

6 Time Series Analysis

Time-series analysis has become a major tool in hydrology. It is used for building mathematical models to generate synthetic hydrologic records, to forecast hydrologic events, to detect trends and shifts in hydrologic records, and to fill in missing data and extend records.

6.1 Hydrologic Time Series

In general, hydrologic processes such as precipitation and runoff evolve on a continuous time scale. For example, a recording gauging station in a stream provides a continuous record of water stage and discharge throughout time. A plot of the flow hydrograph $y(t)$ versus time t constitutes a stream-flow *time series* (see Fig. 6.1a). However, most hydrologic processes of practical interest are defined in a discrete time scale. A discrete time series may be derived by sampling the continuous process at discrete points in time or by integrating the continuous time series over successive time intervals (see Fig. 6.1b). For example, a daily stream-flow series may be derived by sampling the flow of a stream once daily or by integrating the continuous-flow hydrograph on a daily basis. Most hydrologic series are defined on hourly, daily, weekly, monthly, bimonthly, quarterly, and annual intervals.

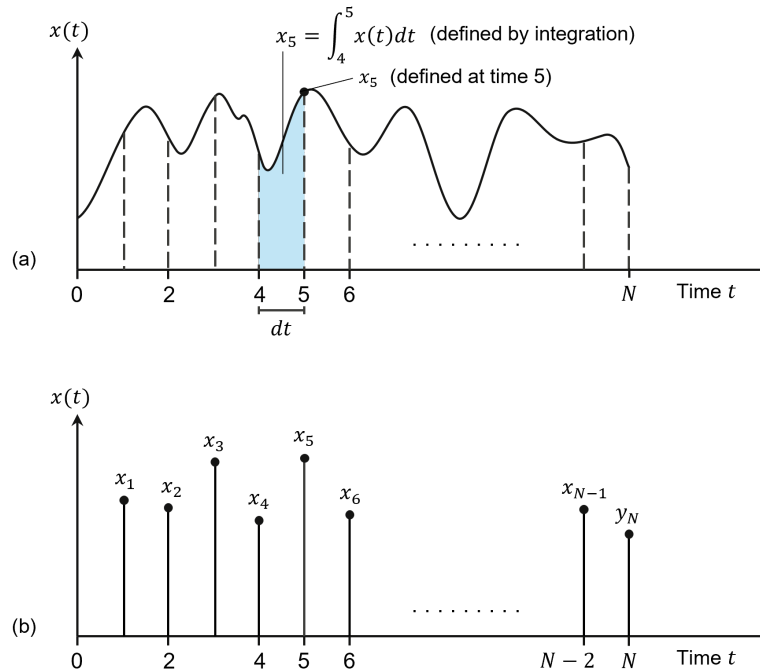


Figure 6.1. A continuous (a) and discrete (b) time series. The discrete time series can be derived by integration over the time interval dt or by taking the observed value at the end of a time interval.

A set of hypothetical replications of the same stochastic process - time series measuring the same process - is called an *ensemble* (see Fig. 6.3). Since in practice we usually only observe one realization of a time series, the ensemble represents all the possible paths the process could have taken under the same underlying probability distribution.

In the framework of stochastic processes, a time series is always made up of a *deterministic* component and a *stochastic component*.

- **Deterministic Component:** The deterministic component can take multiple shapes (see Fig. 6.2):
 - *Trend:* A long smooth movement lasting over the span of the observations. If the movement is described by a straight line, it is a linear trend.
 - *Jump:* A sudden change in the observed magnitudes.
 - *Periodicity:* Oscillatory movement which is repetitive over fixed intervals of time.

- **Stochastic Component:** The stochastic component is constituted by irregular oscillations and random effects which cannot be accounted for deterministically and require probabilistic concepts for description.

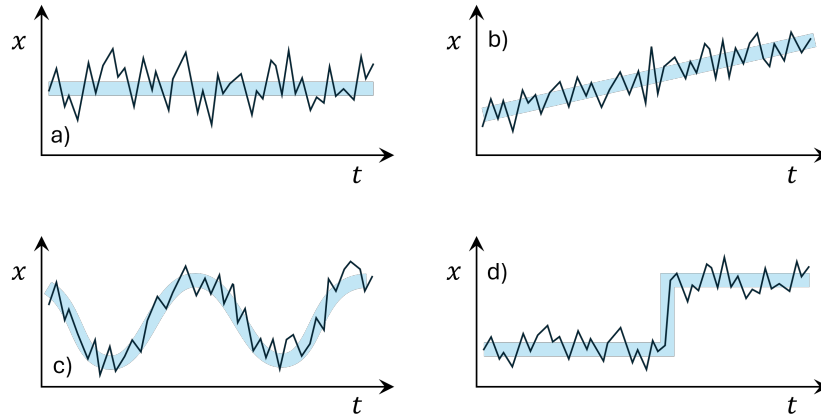


Figure 6.2. Visualization of time series with different deterministic components: a) stochastic component only; b) stochastic component in combination with a linear trend; c) stochastic component in combination with periodicity; and d) stochastic component with a jump.

6.1.1 Stationarity

A stochastic process is said to be *strictly stationary* if the probability density function associated with n observations $(x_1, x_2, \dots, x_n)_t$ made at time origin t is the same as the probability density function associated with n observations $(x_1, x_2, \dots, x_n)_{t+k}$ made at time $t + k$ for all values of n, t and k .

If the statistical properties of a time series do not change over time, the time series is stationary. If the statistical properties do change over time, the time series is considered non-stationary.

Usually, the 2nd order stationarity or *weak stationarity* is tested for, meaning that the mean and covariance of the data does not change over time.

6.1.2 Ergodicity

A time series is one realization of a process, also called a *occurrence* over a finite time interval. The *ensemble* therefore describes a set of realizations of the same process. If the statistical properties of the ensemble are equal to the ones of a single realization, the process is said to be ergodic.

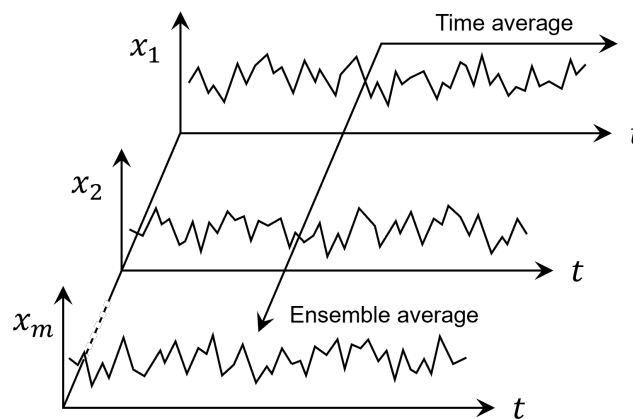


Figure 6.3. Visualization of a set of realizations $x_1, x_2, \dots, x_m$ of the same stochastic process, representing an ensemble.

At each point in time (or location), the time- (or location-) dependent statistics can be computed:

- Mean:

$$\overline{x(t)} = \mathbb{E}[x(t)] = \int_{-\infty}^{\infty} x(t) f_X(x; t) dx = \mu_X(t) \quad (6.1)$$

- Variance:

$$\sigma_X^2(t) = \mathbb{E}[(x(t) - \mu_X(t))^2] = \int_{-\infty}^{\infty} (x(t) - \overline{x(t)})^2 f_X(x; t) dx \quad (6.2)$$

- K-log Covariance: for $k \geq 0$

$$\gamma_k = \text{Cov}[x(t), x(t+k)] = \mathbb{E}[(x(t) - \mu)(x(t+k) - \mu)] \quad (6.3)$$

$$\text{If } k = 0 \quad \rightarrow \quad \gamma_0 = \sigma_X^2$$

- Correlation Coefficient:

$$\rho_k = \frac{\gamma_k}{\gamma_0} \quad (6.4)$$

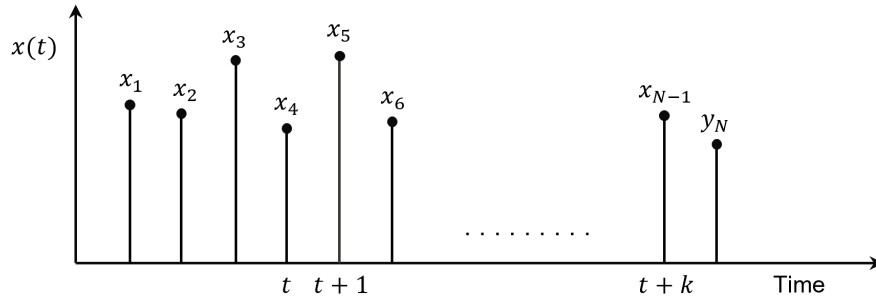


Figure 6.4. Time series with visualization of data points of different lag k in relation to time t .

6.2 Determination and Testing for Trends

6.2.1 Mann-Kendall Test for Trends - Nonparametric

This is a nonparametric test which tests for a trend in a time series without specifying whether the trend is linear or nonlinear. Consider the annual time series x_t with $t = 1, \dots, n$.

H_0 : No Trend - $\mathbb{P}(x_j > x_i) = 0.5$

H_1 : Trend Exists - $\mathbb{P}(x_j > x_i) \neq 0.5$

This test assumes no serial correlation for the p -values to be correct.

Computational steps: Given the time series x_t with $t = 1, \dots, n$

1. Each value $x_{t'}$ ($t = 1, \dots, n$) is compared to all the subsequent values x_t ($t = t' + 1, t' + 2, \dots, n$)
2. A new series z_k is generated:

$$\begin{aligned} z_k &= 1 && \text{if } x_t > x_{t'} \\ z_k &= 0 && \text{if } x_t = x_{t'} \\ z_k &= -1 && \text{if } x_t < x_{t'} \end{aligned}$$

k runs from 1 to $n(n-1)/2$, which is the number of possible comparisons among n data pairs.

3. Compute the S as the sum of all values of z_k :

$$S = \sum_{t'=1}^{n-1} \sum_{t=t'+1}^n z_k \quad (6.5)$$

4. S is modified to be approximated by a normal distribution to describe the large sample approximation:

$$\sigma_S = \sqrt{\frac{n}{18}(n-1)(2n+S)} \quad (6.6)$$

$$z_S = \begin{cases} \frac{S-1}{\sigma_S} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S+1}{\sigma_S} & \text{if } S < 0 \end{cases} \quad (6.7)$$

5. Decision rule:

$$\text{Reject } H_0 \text{ if: } |z_S| > z_{1-\alpha/2} \quad (6.8)$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ point on the standard normal distribution. A lookup table of the standard normal distribution is provided in Table A.1 in the Appendix.

In case of ties, the large sample approximation can be adjusted:

$$\sigma_S = \sqrt{\frac{n}{18}(n-1)(2n+S) - \frac{1}{18} \sum_{i=1}^m e_i(e_i-1)(2e_i+5)} \quad (6.9)$$

where m is the number of tied groups and e_i is the number of observations in the i^{th} tied group.

For $n \leq 10$, an exact form of the decision rule exists. Decision rule:

$$\text{Reject } H_0 \text{ if: } |S| > x_{1-\alpha/2} \quad (6.10)$$

where the critical test statistic $x_{1-\alpha/2}$ is provided in Figure A.1 in the Appendix.

6.2.2 Kendall's Correlation Coefficient

Kendall's τ quantifies the strength of a monotonic relationship between two variables. Its value ranges between -1 (perfect negative correlation) and +1 (perfect positive correlation).

$$\tau = \frac{S}{n(n-1)/2} \quad (6.11)$$

with S as defined in Equation 6.5 above.

6.2.3 Linear Regression - Parametric

Assume that y is the variable of interest or response variable, referred to as the *dependent variable*, and x is the *explanatory variable*, which can be a random variable or a measure of time, downstream coordinate, etc.

A simple linear trend can then be written as

$$y = \beta_1 \cdot x + \beta_0 \quad (6.12)$$

where β_0 represents the intercept and β_1 represents the slope or the linear trend.

The ordinary least square linear regression can be expressed as

$$y_i = \beta_0 + \beta_1 \cdot x_i + \varepsilon_i \quad (6.13)$$

for all data pairs $i = 1, \dots, n$ with sample size n and residuals ε_i .

Procedure to develop and test a regression model:

1. Plot the data (y vs. x) on a scatterplot and check for two characteristics:
 - a. Is the relationship linear or curved?

- b Is the variability of y different for different levels of x ?
 Constant variance $\rightarrow$ relationship is *homoscedastic*
 Non-constant variance $\rightarrow$ relationship is *heteroscedastic*
- c If the data is heteroscedastic or curved, the data should be transformed according to the table below.

Appearance	Linear	Curved
Homoscedastic	Proceed with linear regression	Transform x to achieve linearity.
Heteroscedastic	Transform y and possibly x to achieve homoscedasticity.	Transform y and possibly x to achieve homoscedasticity and linearity.

When the relationship between the variables on the x - and y -axis do not describe a linear relationship, mathematical transformations can be employed to straighten the dependence. As visualized in Figure 6.5, different curvature shapes necessitate different transformations of the x and y variables. Moving up on the ladder of transformations means applying a convex transformation (e.g., x^2 , x^3 , x^4 , ...) and conversely, moving down means applying a concave transformation (e.g., $\sqrt{x}$, $\log x$, $1/x$, $1/x^2$, $1/x^3$, ...). If a given transformation is not sufficient to achieve a linear relation, move further along the ladder of transformations until a linear relationship is reached.

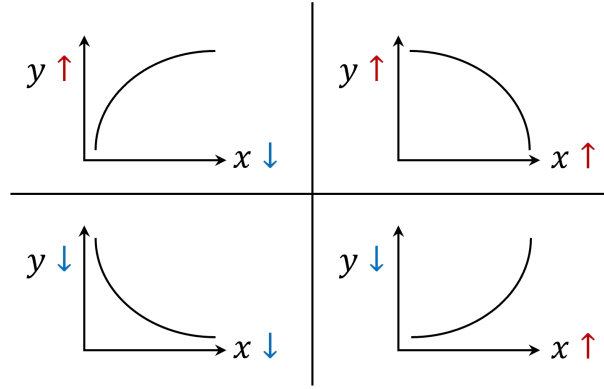


Figure 6.5. Options for mathematical transformations to remove a curvature from the relationship between a x and y variable. The up $\uparrow$ and down $\downarrow$ thereby represent moving up or down the ladder of transformations as described above.

2. Estimate the parameter of the model β_0 (intercept) and β_1 (slope). The parameters are found based on the minimization of the objective function Q described below.

$$(\hat{\beta}_0, \hat{\beta}_1) = \operatorname{argmin}\{Q(\beta_0, \beta_1)\} \quad (6.14)$$

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n \hat{\varepsilon}_i^2 \quad (6.15)$$

$$= \sum_{i=1}^n (y_i - \beta_0 - \beta_1 \cdot x_i)^2 \quad (6.16)$$

$$= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (6.17)$$

By expansion to get the quadratic expansion in β_0 and β_1 , we can derive minimizing values of the function arguments:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \cdot \bar{x} \quad (6.18)$$

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} \quad (6.19)$$

where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{mean of } x \quad (6.20)$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad \text{mean of } y \quad (6.21)$$

$$S_{YY} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n(\bar{y})^2 \quad \text{sum of squares } y \quad (6.22)$$

$$S_{XX} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n(\bar{x})^2 \quad \text{sum of squares } x \quad (6.23)$$

$$S_{XY} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i \cdot y_i) - n \bar{x} \bar{y} \quad \text{sum of } x\text{-}y \text{ cross product} \quad (6.24)$$

3. Use the t -statistic to determine the significance of the slope $\hat{\beta}_1$ (and intercept $\hat{\beta}_0$).

$$H_0 : \hat{\beta}_1 = 0$$

$$H_1 : \hat{\beta}_1 \neq 0 \quad \text{Statistically significant linear relationship between } x \text{ and } y.$$

Test statistic t :

$$t = \frac{\hat{\beta}_1}{S/\sqrt{S_{XX}}} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \quad (6.25)$$

where

$$S^2 = \frac{S_{YY} - \hat{\beta}_1 \cdot S_{XY}}{n-2} = \frac{\sum_{i=1}^n e_i^2}{n-2} \quad \text{Mean Square Error (MSE)} \quad (6.26)$$

$$e_i = y_i - \hat{y}_i \quad \text{Residual} \quad (6.27)$$

$$S = \sqrt{S^2} \quad \text{Standard error of regression} \quad (6.28)$$

$$r = \frac{S_{XY}}{\sqrt{S_{XX} \cdot S_{YY}}} = \hat{\beta}_1 \sqrt{\frac{S_{XX}}{S_{YY}}} \quad \text{Correlation coefficient} \quad (6.29)$$

Decision rule:

$$\text{Reject } H_0 \text{ if: } |t| > t_{1-\alpha/2, n-2}$$

where $t_{1-\alpha/2, n-2}$ is the $1 - \alpha/2$ quantile of a Student's t distribution with $n - 2$ degrees of freedom. A lookup table of the Student's t distribution can be found in Table A.2 in the Appendix.

4. Analyze the residuals e_i :

$$e_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 \cdot x_i) \quad (6.30)$$

Three types of residual plots are useful:

- Scatterplot of the residuals e_i vs. predicted $\hat{y}_i$. Curvature or heteroscedasticity indicate a problem with the model choice.
- Plot the residuals vs. time or the appropriate 1D or 2D space coordinate (if there is a time or space order of the observations). A good residual pattern should look like random noise.
- Check the normality of the residuals on a probability plot.

The residuals can also be plotted against other explanatory variables to determine whether they should be included in a multiple regression.

Example - Mathematical Transformation to Achieve Linearity:

Figure 6.6 on the left shows measurements of discharge Q in relation to the concentration of total dissolved solids, with a clearly visible nonlinearity being present. Following the instructions outlined above, this can be transformed into a linear relationship by transforming either variable down the ladder of transformations. In this example, we focus on only transforming the variable on the x -axis - discharge Q . As seen when looking at Figure 6.6 on the right, starting with taking the root and plotting $\sqrt{Q}$ against TSD still preserves some nonlinearity.

Figure 6.7 shows two further transformation options: $\log Q$ (figures on the left) and $1/Q$ (figures on the right). As we can see when looking at the top images, both cases visibly present a more linear behavior. The residuals (at the top in Fig. 6.8) for both transformation approaches present no obvious striking features, indicating that both could be a good fit for this data set.

If the linear model describes the data well, we can assume the residuals ($TDS_{obs} - TDS_{model}$) to be normally distributed. The distribution of the residuals assuming a normal distribution are shown in the bottom row of Figure 6.8. In both transformation approaches the linear model underestimates the TDS concentration for the smallest and largest observations. Looking at the PPCC r correlation, we can see that the fit is higher ($r = 0.9825$) for the $1/Q$ -transformation compared to the $\log Q$ -transformation ($r = 0.9752$). This indicates that for this data set, transforming the discharge measurements to $1/Q$ achieves the best linear fit out of the tested transformations.

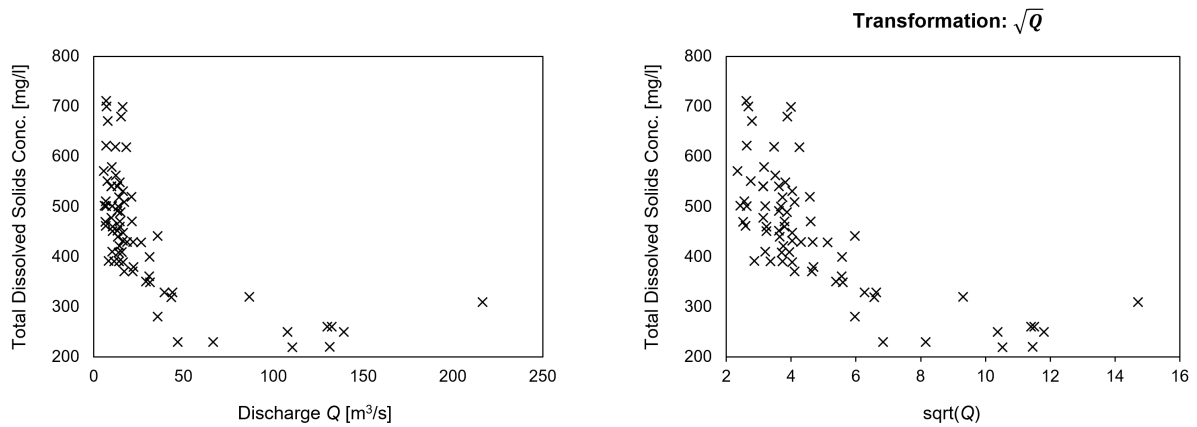


Figure 6.6

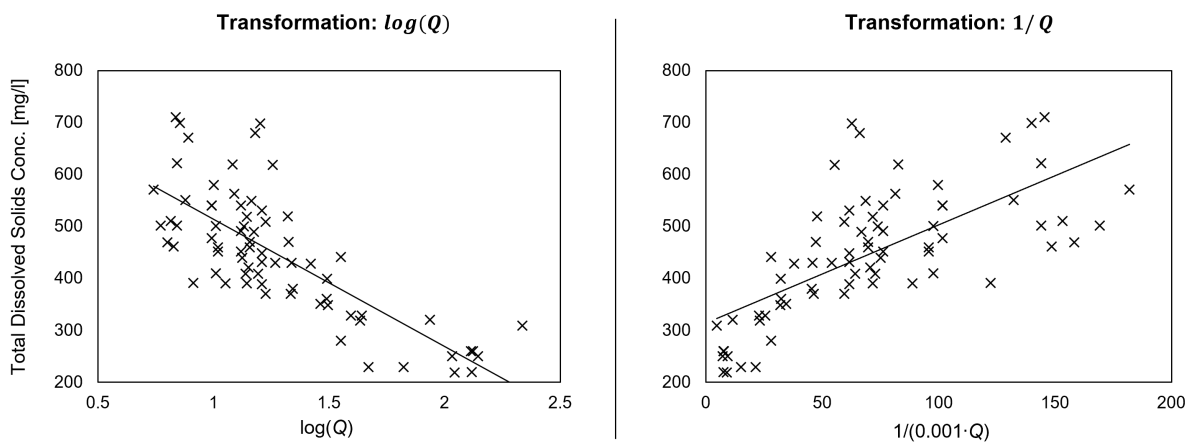


Figure 6.7

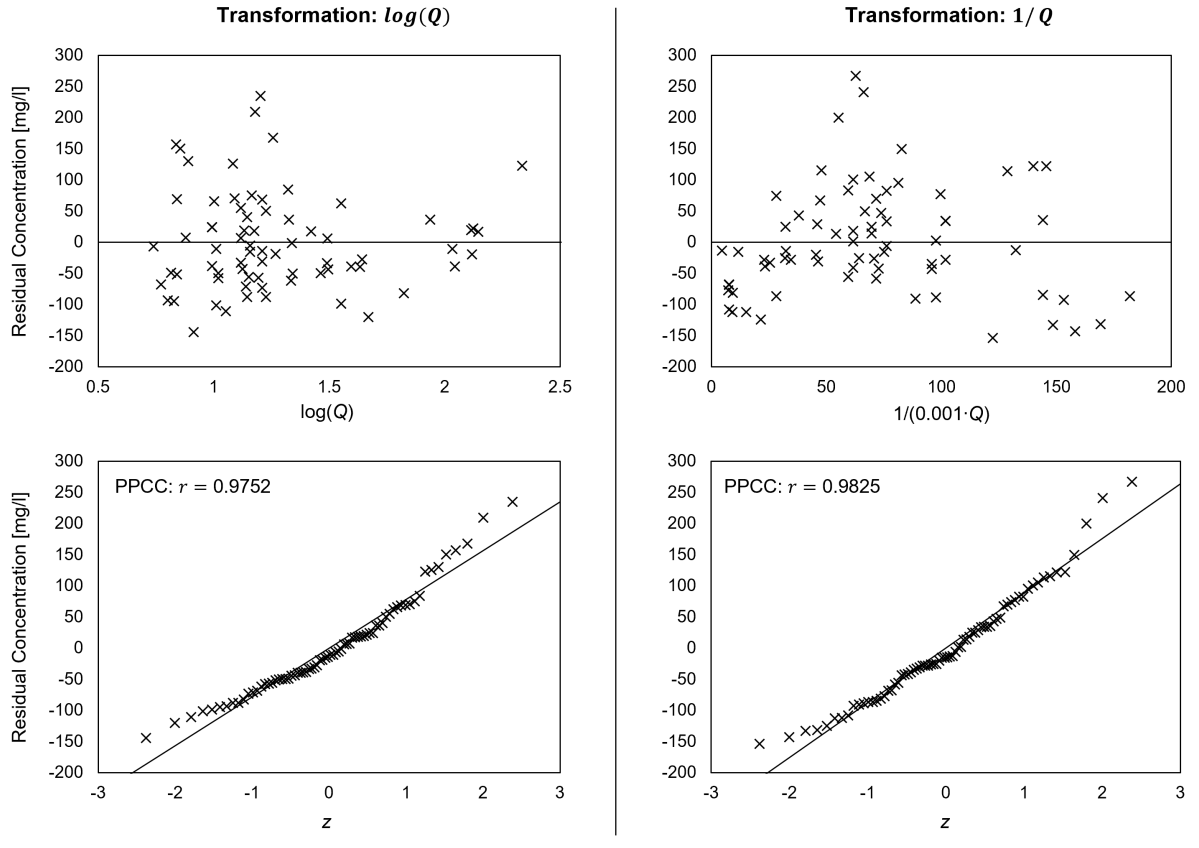


Figure 6.8

6.2.4 Kendall-Theil Linear Robust Line - Nonparametric

This test does not depend on the assumption of normality of the residuals and is not as strongly affected by outliers as the linear regression.

H_0 : No Trend - slope $\beta_1 = 0$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot x \quad (6.31)$$

Computational steps:

1. Each data pair (x, y) is compared to all other pairs for a total of $n(n-1)/2$ possible combinations.
2. For each of these comparisons, a slope $\Delta y / \Delta x$ is computed.
3. The median of all possible pairwise slopes is the nonparametric estimate of the slope $\hat{\beta}_1$:

$$\hat{\beta}_1 = \text{median} \left(\frac{y_j - y_i}{x_j - x_i} \right) \quad (6.32)$$

for all $i < j$ with $i = 1, 2, \dots, n-1$ and $j = 2, 3, \dots, n$.

The intercept is given by

$$\hat{\beta}_0 = y_{\text{med}} - \hat{\beta}_1 \cdot x_{\text{med}} \quad (6.33)$$

where x_{med} and y_{med} represent the median of x and y , respectively.

6.3 Determination and Testing for Shifts

Tests used to compare two non-overlapping sets of data can be used to identify shifts (e.g., before and after a point in time). They are identical to the ones previously described for comparison of two samples:

- Rank-sum test - nonparametric
- Two sample t -test - parametric

When to use shift tests:

- The record being analyzed is naturally broken into two distinct time periods with a gap between.
- A known event has occurred at a specific point in time during the record which likely changed the record.

Note: Do not assume a shift without one of the circumstances described above applying.

6.4 Testing for Seasonality / Periodicity

Testing for seasonality is mostly done by plotting the statistic under consideration versus periodicity causing variable (e.g., month or season). There exist procedures for testing for seasonality (e.g., ANOVA or Kruskal-Wallis test).

6.5 Removal of Nonstationarities

6.5.1 Removal of Trends

Trends can be removed by several methods:

- Regression against the trend-causing variable
- Mathematical transformation
- Standardization to get the stationary time series z_t :
 - Removal of the linear trend $\bar{x}_t$ (see Fig. 6.9)

$$z_t = x_t - \bar{x}_t \quad (6.34)$$

- If the standard deviation is also a function of time:

$$z_t = \frac{x_t - \bar{x}_t}{S_t} \quad (6.35)$$

where S_t is the standard deviation that is subject to change over time t .

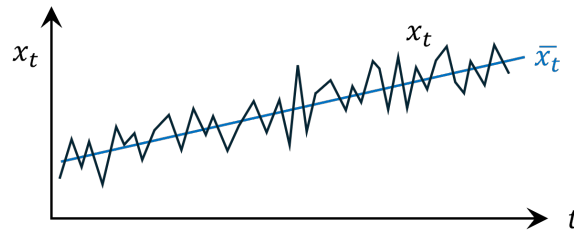


Figure 6.9. Visualization of a timeseries x_t with a linear trend $\bar{x}_t$ represented by the blue line.

6.5.2 Removal of Shifts

To remove a shift from your data, employ a standardization to get the stationary time series z_t

$$z_t = x_t - \bar{x}_t \quad (6.36)$$

where $\bar{x}_t$ represents the mean of the differing segments.

If there is also a shift in standard deviation present, employ:

$$z_t = \frac{x_t - \bar{x}_t}{S_t} \quad (6.37)$$

where S_t is the standard deviation of the different segments before and after the shift.

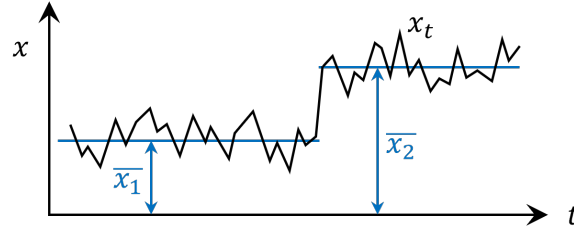


Figure 6.10. Visualization of a time series x_t with a shift, and mean $\bar{x}_1$ and $\bar{x}_2$ before and after the shift.

6.5.3 Removal of Periodicity

To remove periodicity from your data, there are two options:

- *Standardization:* Subtract the seasonal mean $\bar{x}_t$ and divide by the seasonal standard deviation S_t as described by Equation 6.37.
- *Spectral Analysis:* Identify the fundamental period and subharmonics sufficient to describe the periodic movement.

7 Stochastic Models

Stochastic models can be applicable for many hydrologic processes, particularly stream-flow and precipitation processes. The idea is to construct a linear model that adequately describes the autocorrelation structure of the data. The presented models belong to the class of autoregressive (AR) and autoregressive with moving average term (ARMA) modeling schemes.

7.1 Autocorrelation

The autocorrelation coefficient of order k is described by

$$\rho_k = \frac{\text{COV}(x_t, x_{t+k})}{\sigma_X^2} \quad (7.1)$$

From data, we describe autocorrelation coefficient r_k as the linear dependence between observations k units apart:

$$r_k = \frac{c_k}{c_0} \quad (7.2)$$

$$c_k = \frac{1}{n} \sum_{t=1}^{n-k} (x_{t+k} - \bar{x})(x_t - \bar{x}) \quad \text{for } k \geq 0 \quad (7.3)$$

Consider a stationary time series x_t . If the value of x at time t depends on the values of x at the previous times ($t-1, t-2, \dots$), we say that the series has *memory*. Alternatively, if x at time t does not depend on previous values, then the series is *uncorrelated* or *independent*.

Figure 7.1 visualizes the autocorrelation of annual streamflow at three different location in the Nile river catchment, showing the longest memory in sample (a) and the shortest memory in sample (c).

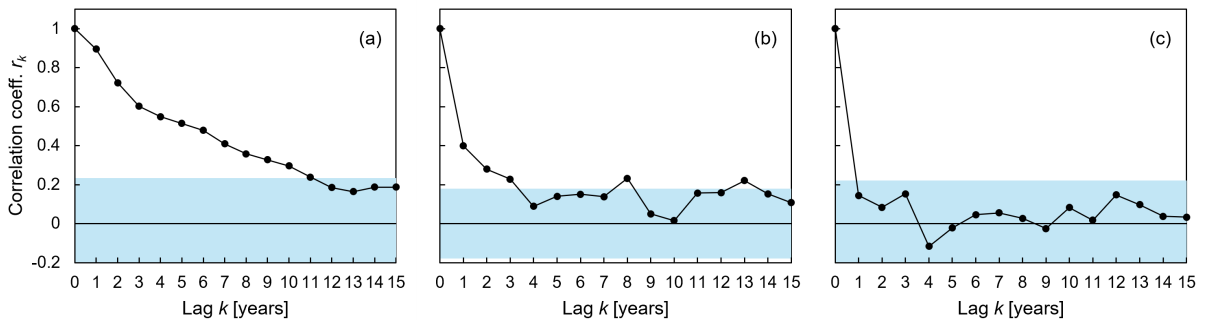


Figure 7.1. Correlogram of annual flows of (a) the White Nile River at Mongalla (1914-1983), (b) the Nile River at Aswan (1871-1989), and (c) the Blue Nile River at Khartoum (1912-1989). The shaded area in blue represents the 95 % confidence limits. (Adapted from Maidment 1992, Fig. 19.2.2)

The *confidence limits* on the correlogram are built assuming that r_k follows a normal distribution $\mathcal{N}(0, 1/n)$:

$$r_{k, 1-\alpha} = \frac{-1 \pm z_{1-\alpha/2} \sqrt{n-k-1}}{n-k} \simeq \pm \frac{z_{1-\alpha/2}}{\sqrt{n}} \quad (7.4)$$

where n represents the number of observations and $z_{1-\alpha/2}$ is the $1 - \alpha/2$ point on the standard normal distribution. A lookup table of the standard normal distribution is provided in Table A.1 in the Appendix.

The 95 % confidence interval can be approximated as

$$r_{0.95} \simeq \pm \frac{1.96}{\sqrt{n}} \quad (7.5)$$

7.2 Steps in Setting up a Stochastic Model

To set up a stochastic model, the following three steps have to be followed:

1. *Identification*: Use the data to understand what class of models is most appropriate by analyzing the autocorrelation function ACF and the partial autocorrelation function PACF.
2. *Estimation*: Use the data to estimate the parameters of the model.
3. *Analysis of the residuals*: If the model is a good fit for the data, the residuals should be white noise. Validate this by checking them for normality and independence.

7.3 Autoregressive AR Models

For a stationary time series $z_t = x_t - \bar{x}_t$, the autoregressive model of order p is described by

$$z_t = \phi_1 \cdot z_{t-1} + \phi_2 \cdot z_{t-2} + \dots + \phi_p \cdot z_{t-p} + a_t \quad (7.6)$$

$$= \left[\sum_{i=1}^p \phi_i \cdot z_{t-i} \right] + a_t \quad (7.7)$$

where a_t represents the uncorrelated and normally distributed residuals $\sim \mathcal{N}(0, \sigma_a^2)$.

Autocorrelation function ACF of the AR(p) model: The covariance γ_k at lag k is described by

$$\gamma_k = \mathbb{E}[z_t \cdot z_{t-k}] \quad (7.8)$$

$$= \mathbb{E}[\phi_1 \cdot z_{t-1} \cdot z_{t-k} + \phi_2 \cdot z_{t-2} \cdot z_{t-k} + \dots + \phi_p \cdot z_{t-p} \cdot z_{t-k} + a_t \cdot z_{t-k}] \quad (7.9)$$

There is no covariance between the noise and z_{t-k} :

$$\mathbb{E}[z_{t-k} \cdot a_t] = 0 \quad (7.10)$$

We can therefore simplify to

$$\gamma_k = \phi_1 \cdot \gamma_{k-1} + \phi_2 \cdot \gamma_{k-2} + \dots + \phi_p \cdot \gamma_{k-p} \quad (7.11)$$

Dividing by the variance $\gamma_0 = \mathbb{E}[z_t \cdot z_t]$ results in

$$\rho_k = \phi_1 \cdot \rho_{k-1} + \phi_2 \cdot \rho_{k-2} + \dots + \phi_p \cdot \rho_{k-p} \quad (7.12)$$

Note: It holds that $\rho_k = \rho_{-k}$

By writing the expression of ρ_k for $k = 1, \dots, p$, we obtain the *Yule-Walker equations*:

$$\begin{cases} \rho_1 = \phi_1 \cdot \rho_0 + \phi_2 \cdot \rho_1 + \phi_3 \cdot \rho_2 + \dots + \phi_p \cdot \rho_{p-1} \\ \rho_2 = \phi_1 \cdot \rho_1 + \phi_2 \cdot \rho_2 + \phi_3 \cdot \rho_1 + \dots + \phi_p \cdot \rho_{p-2} \\ \rho_3 = \phi_1 \cdot \rho_2 + \phi_2 \cdot \rho_1 + \phi_3 \cdot \rho_3 + \phi_4 \cdot \rho_1 + \dots + \phi_p \cdot \rho_{p-3} \\ \vdots \\ \rho_p = \phi_1 \cdot \rho_{p-1} + \phi_2 \cdot \rho_{p-2} + \phi_3 \cdot \rho_{p-3} + \dots + \phi_p \cdot \rho_p \end{cases} \quad (7.13)$$

The parameters $\phi_1, \dots, \phi_p$ of the model are found by setting ρ_k equal to the estimated autocorrelation coefficient r_k .

The variance σ_a^2 is estimated from

$$\sigma_z^2 = \mathbb{E}[z_t \cdot z_t] \quad (7.14)$$

$$= \gamma_0 \quad (7.15)$$

$$= \phi_1 \cdot \gamma_1 + \phi_2 \cdot \gamma_2 + \dots + \phi_p \cdot \gamma_p + \sigma_a^2 \quad (7.16)$$

with $\sigma_a^2 = \mathbb{E}[a_t \cdot a_t]$.

Dividing by $\sigma_z^2 = \gamma_0$ leads to

$$\sigma_z^2 = \frac{\sigma_a^2}{1 - \phi_1 \cdot \rho_1 - \phi_2 \cdot \rho_2 - \dots - \phi_p \cdot \rho_p} \quad (7.17)$$

In hydrological applications, usually an autoregressive model of order 1 AR(1) or two AR(2) is sufficient to model the variable of interest.

7.3.1 Autoregressive Model of Order 1 - AR(1)

The model is described by

$$z_t = \phi_1 \cdot z_{t-1} + a_t \quad (7.18)$$

The autocorrelation function ACF of the AR(1) is obtained from the Yule-Walker equations and describes an exponential decay:

$$\left\{ \begin{array}{lll} \rho_1 & = & \phi_1 \\ \rho_2 & = & \phi_1 \cdot \rho_1 = \rho_1^2 \\ \rho_3 & = & \phi_1 \cdot \rho_2 = \phi_1^2 \cdot \rho_1 = \rho_1^3 \\ & \vdots & \\ \rho_k & = & \phi_1^k = \rho_1^k \end{array} \right. \quad (7.19)$$

Under stationary conditions, it applies that $|\phi_1| < 1$.

The variance of the noise σ_a^2 is given by

$$\sigma_z^2 = \frac{\sigma_a^2}{1 - \phi_1 \cdot \rho_1} \quad (7.20)$$

$$= \frac{\sigma_a^2}{1 - \rho_1^2} \quad (7.21)$$

7.3.2 Autoregressive Model of Order 2 - AR(2)

The model is described by

$$z_t = \phi_1 \cdot z_{t-1} + \phi_2 \cdot z_{t-2} + a_t \quad (7.22)$$

The parameters ϕ_1 and ϕ_2 are estimated from the Yule-Walker equation:

$$\begin{array}{lll} \rho_1 & = & \phi_1 + \phi_2 \cdot \rho_1 \\ \rho_2 & = & \phi_1 \cdot \rho_1 + \phi_2 \end{array} \quad \Rightarrow \quad \begin{array}{ll} \phi_1 & = \frac{\rho_1(1 - \rho_2)}{1 - \rho_1^2} \\ \phi_2 & = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} \end{array} \quad (7.23)$$

Find the empirical autocorrelation for the first two lags r_1 and r_2 and substitute them for ρ_1 and ρ_2 , respectively. With that, you find the estimated parameters of your model $\hat{\phi}_1$ and $\hat{\phi}_2$.

The variance of the noise σ_a^2 is given by

$$\sigma_z^2 = \frac{\sigma_a^2}{1 - \phi_1 \cdot \rho_1 - \phi_2 \cdot \rho_2} \quad (7.24)$$

The autocorrelation function ACF of the AR(2) model is

$$\rho_k = \phi_1 \cdot \rho_{k-1} + \phi_2 \cdot \rho_{k-2} \quad \text{for } k > 1 \quad (7.25)$$

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} \quad \text{for } k = 1 \quad (7.26)$$

The stationarity condition implies that

$$\begin{aligned} \phi_1 + \phi_2 &< 1 & -1 < \rho_1 < 1 \\ \phi_2 - \phi_1 &< 1 & \implies -1 < \rho_2 < 1 \\ -1 < \phi_2 < 1 & & \rho_1^2 < \frac{1}{2}(\rho_2 + 1) \end{aligned} \quad (7.27)$$

7.4 Moving Average MA Model

An MA model expresses the current value of a time series as a linear function of current and past random shocks. The moving average model MA(q) of order q is described as

$$z_t = a_t - \theta_1 \cdot a_{t-1} - \theta_2 \cdot a_{t-2} - \cdots - \theta_q \cdot a_{t-q} \quad (7.28)$$

The autocorrelation function ACF of the moving average model MA(q) cuts off at lag $q + 1$, will all subsequent values $\rho_k = 0$ for $k > q$.

7.4.1 Moving Average Model of Order 1 - MA(1)

The model is described by

$$z_t = a_t - \theta_1 \cdot a_{t-1} \quad (7.29)$$

The invertibility condition implies that

$$|\theta_1| < 1 \quad (7.30)$$

The variance of the noise is given by

$$\sigma_z = (1 + \theta_1^2) \cdot \sigma_a^2 \quad (7.31)$$

The autocorrelation function ACF of the MA(1) model is

$$\rho_k = \begin{cases} \frac{-\theta_1}{1 + \theta_1^2} & \text{for } k = 1 \\ 0 & \text{for } k \geq 2 \end{cases} \quad (7.32)$$

7.4.2 Moving Average Model of Order 2 - MA(2)

The model is described by

$$z_t = a_t - \theta_1 \cdot a_{t-1} - \theta_2 \cdot a_{t-2} \quad (7.33)$$

The invertibility conditions imply

$$\theta_1 + \theta_2 < 1 \quad (7.34)$$

$$\theta_2 - \theta_1 < 1 \quad (7.35)$$

$$-1 < \theta_2 < 1 \quad (7.36)$$

The variance of the noise is given by

$$\sigma_z = (1 + \theta_1^2 + \theta_2^2) \cdot \sigma_a^2 \quad (7.37)$$

The autocorrelation function ACF of the MA(2) model is

$$\begin{cases} \rho_1 = \frac{-\theta_1(1 - \theta_2)}{1 + \theta_1^2 + \theta_2^2} \\ \rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2} \\ \rho_k = 0 & \text{for } k \geq 3 \end{cases} \quad (7.38)$$

7.5 Autoregressive Moving Average ARMA Model

The ARMA(p, q) model of autoregressive order p and moving average order q is described by

$$z_t = \phi_1 \cdot z_{t-1} + \dots + \phi_p \cdot z_{t-p} + a_t - \theta_1 \cdot a_{t-1} - \dots - \theta_q \cdot a_{t-q} \quad (7.39)$$

The autocorrelation function ACF has a mixed behavior:

- for $k \leq q$: γ_k depends on ϕ and θ
- for $k > q$: ACF of AR(p)

7.5.1 Autoregressive Moving Average Model of Order (1,1) - ARMA(1,1)

The ARMA(1,1) model is described by

$$z_t = \phi_1 \cdot z_{t-1} + a_t - \theta_1 \cdot a_{t-1} \quad (7.40)$$

The stationarity and invertibility conditions imply

$$\begin{aligned} -1 < \phi_1 < 1 \\ -1 < \theta_1 < 1 \end{aligned} \implies \begin{aligned} |\rho_2| < |\rho_1| \\ \rho_2 > \rho_1(2\rho_1 + 1) \quad \text{for } \rho_1 < 0 \\ \rho_2 > \rho_1(2\rho_1 - 1) \quad \text{for } \rho_1 > 0 \end{aligned} \quad (7.41)$$

The autocorrelation function of the ARMA(1,1) model is given by

$$\begin{cases} \rho_1 = \frac{(1 - \phi_1\theta_1)(\phi_1 - \theta_1)}{1 - 2\phi_1\theta_1 + \theta_1^2} \\ \rho_k = \phi_1 \cdot \rho_{k-1} \end{cases} \quad \text{for } k > 1 \quad (7.42)$$

The variance of the noise is given by

$$\sigma_z^2 = \frac{1 + \theta_1^2 - 2\phi_1\theta_1}{1 - \phi_1^2} \cdot \sigma_a^2 \quad (7.43)$$

7.6 Partial Autocorrelation Function PACF

The partial autocorrelation $\phi_{k,k}$ is estimated by fitting successive autoregressive processes of increasing order k . $\phi_{k,k}$ is the last coefficient in the AR model of order k , where the first k in the subscript represents the order of the model and the second k in the subscript indicates the coefficient.

$$k = 1 : \quad z_t = \phi_{1,1} \cdot z_{t-1} + a_t \quad \rightarrow \phi_{1,1} \quad (7.44)$$

$$k = 2 : \quad z_t = \phi_{2,1} \cdot z_{t-1} + \phi_{2,2} \cdot z_{t-2} + a_t \quad \rightarrow \phi_{2,2} \quad (7.45)$$

$\vdots$

The coefficients are estimated using the Yule-Walker equations:

$$\hat{\phi}_{k,k} = \frac{r_k - \sum_{j=1}^{k-1} \hat{\phi}_{k-1,j} \cdot r_{k-j}}{1 - \sum_{j=1}^{k-1} \hat{\phi}_{k-1,j} \cdot r_j} \quad (7.46)$$

where

$$\hat{\phi}_{k,j} = \hat{\phi}_{k-1,j} - \hat{\phi}_{k,k} \cdot \hat{\phi}_{k-1,k-j} \quad \text{for } j = 1, \dots, k-1 \quad (7.47)$$

Table 7.1. Behavior of the autocorrelation function ACF and the partial autocorrelation function PACF for different models (autoregressive AR, moving average MA and autoregressive moving average ARMA).

Process	ACF	PACF
AR(p)	$\rho_k = \sum_{j=1}^p \phi_j \cdot \rho_{k-j}$	Cuts off at lags $> p$
MA(q)	Cuts off at lags $> q$	Gradually decays
ARMA(p, q)	Irregular for lags $\leq q$ AR(p) type for lags $> q$	Decays

Example - Model Order Choice 1:

Figure 7.2 (a) and (b) show the autocorrelation function ACF and partial autocorrelation function PACF, respectively, for a given timeseries of observations. Figure 7.2 (c) shows the ACF for the residuals of different model choices.

(a) and (b) show a autocorrelation $r_k \rightarrow 0$ for $k > 2$ and $\hat{\phi}_{k,k} \neq 0$ for $k = 1, 2$. As outlined in Table 7.1 above, we expect the PACF to be insignificant for lags $> p$ of an AR(p) model. Based on the observed partial autocorrelation $\hat{\phi}_{k,k}$, this indicates that an autoregressive model of up to order 2 is possible, i.e. an AR(1) or AR(2).

For any adequate model, the ACF or the residuals should be uncorrelated. As shown in (c), the ACF r_k falls barely on the confidence bounds for $k = 1$ and outside for $k = 2$, indicating that the model AR(1) is not sufficient to reproduce the memory adequately. The AR(2) model on the other hand falls within the confidence bounds starting at lag $k = 1$.

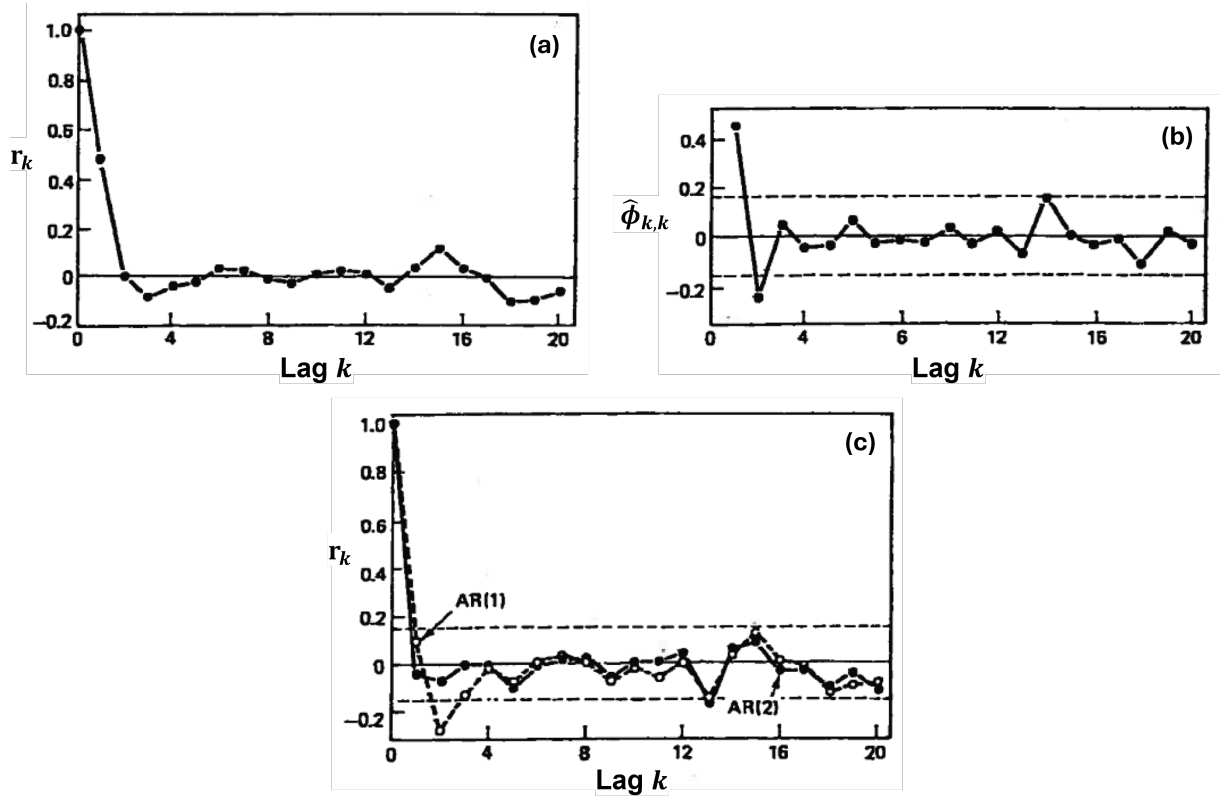


Figure 7.2

Example - Model Order Choice 2:

Figure 7.3 similarly shows the ACF and PACF of a second timeseries of observations in (a) and (b), respectively, and the ACF of the residuals for different models in (c).

The PACF $\hat{\phi}_{k,k}$ only shows a significant value for $k = 1$, sitting at or within the confidence bounds for lags $k > 1$. This indicates that a model of order 1 AR(1) might be enough. This is supported when looking at the shape of the ACF, which approximates an exponential decay as expected from the AR(1) model as described in the chapter above.

Both an AR(1) and AR(3) model were fit to the data and the autocorrelation of the residuals is displayed in (c). We can see that the ACF under both models is within the confidence bounds for all lags $k \geq 1$. This indicates that an AR(1) model is adequate for this application and the additional complexity (and therefore computational power) added by the AR(3) is not necessary.

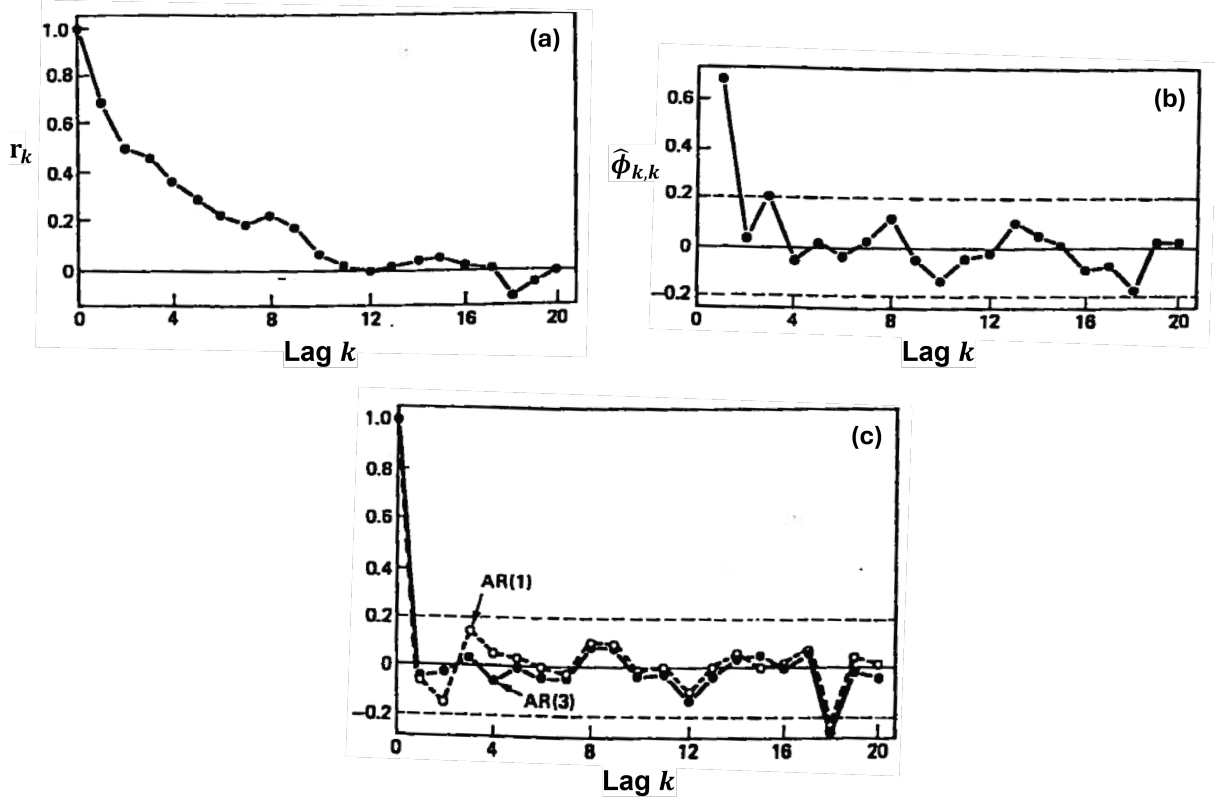


Figure 7.3

7.7 Verification of the Model Choice

To validate the model choice, test for two characteristics:

- Test the independence and normality of the residuals.
- Compare the sample correlogram with the model correlogram.

For the AR(1) model, the residuals are computed by rearranging the equation to obtain a_t as a function of observations z_t and z_{t-1} , and the estimated parameter $\hat{\phi}_1$ as shown in Table 7.2.

Table 7.2. Computation of the residual of the AR(1) model with $z_t = \hat{\phi}_1 \cdot z_{t-1} + a_t$.

Observations	Model	Residual a_t
z_1		
z_2	$\hat{\phi}_1 \cdot z_1$	$z_2 - \hat{\phi}_1 \cdot z_1$
z_3	$\hat{\phi}_1 \cdot z_2$	$z_3 - \hat{\phi}_1 \cdot z_2$
$\vdots$	$\vdots$	$\vdots$

7.7.1 Test for Independence of Residuals

The independence of the residuals is tested by:

1. Compute and plot the ACF of the residuals and check whether it is within the confidence limits for all lags $k \geq 1$.

2. Porte-Manteau Test:

- H_0 : The residuals $\hat{a}_t$ are independent.
- Test statistic:

$$Q = N \sum_{k=1}^L r_k^2(\hat{a}) \quad (7.48)$$

where $r_k(\hat{a})$ is the autocorrelation coefficient of the residuals, L is the maximum lag and N is the number of observations.

- Q follows a χ^2 distribution with $L - p - q$ degrees of freedom:

$$Q \sim \chi^2_{L-p-q} \quad (7.49)$$

- Decision Rule:

$$\text{Reject } H_0 \text{ if: } Q > \chi^2_{L-p-q} \quad (7.50)$$

7.7.2 Test for Normality of Residuals

Test the residuals $\hat{a}_t$ by looking at the probability plot and performing a PPCC test.

7.8 Generation of Synthetic Series with ARMA Models

Here, we outline the steps to generate synthetic timeseries for an example AR(1) model.

$$z_t = \hat{\phi}_1 \cdot z_{t-1} + a_t \quad \text{where } a_t \sim \mathcal{N}(0, \sigma_a^2) \quad (7.51)$$

1. Start by generating m values from a normal distribution $\mathcal{N}(0, \sigma_a^2)$ to create the values of a_t .
2. Compute the values of z_t as

$$z_t = \hat{\phi}_1 \cdot z_{t-1} + a_t \quad (7.52)$$

where $\hat{\phi}_1$ is the fitted parameter, z_{t-1} is the computed value from the previous time step (choose $z_{t-1} = 0$ for the first iteration), and a_t is the noise generated above.

With this, you obtain a series of m values of z .

3. If the mean or any other instationarity $\bar{x}_t$ was removed before, add it back to obtain the synthetic time series x_t .

$$x_t = \bar{x}_t + z_t \quad (7.53)$$

There are procedures to generate synthetic series that preserve the skewness of the data (e.g. AR Gamma or AR LN) where the noise term a_t carries the distribution.

7.9 Multivariate Models

When dealing with multiple variables measured at one gauging station or the same variable at multiple locations (correlated), a multivariate model can be employed.

The sample lag k now represents the cross-correlation. Given two time series measured at two sites i, j (or two variables) $x_t^{(i)}$ and $x_t^{(j)}$

$$r_k^{(i,j)} = \frac{c_k^{(i,j)}}{[c_0^{(i,i)} \cdot c_0^{(j,j)}]^{1/2}} \quad (7.54)$$

$$c_k^{(i,j)} = \frac{1}{N} \sum_{t=1}^{N-k} (x_{t+k}^{(i)} - \bar{x}^{(i)})(x_t^{(j)} - \bar{x}^{(j)}) \quad (7.55)$$

The plot of $r_k^{(i,j)}$ versus the lag k expresses the cross-correlogram or cross-correlation function.

The ARMA (or AR) models are similar to the univariate models introduced in the previous chapters, but each site uses information from the other sites, which is introduced through the cross-correlation. The equations are therefore expressed in matrix form.

Bibliography

Maidment, David R (1992). *Handbook of hydrology*.

A Appendix

A.1 Tables of Critical Values

Table A.1. Tabled values for the normal standard distribution, relating the curve left of the reduced variate z under the curve. For values with $z < 0$, employ the relation $\Phi(-z) = 1 - \Phi(z)$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997

Table A.2. Table of the Student's t distribution for different degrees of freedom df and significance levels α .

df	α level									
	0.25	0.2	0.15	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
5	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
6	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
7	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
8	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
9	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
10	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
11	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
12	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
13	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
14	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
15	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
16	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
17	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
18	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
19	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
20	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
21	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
22	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
23	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
24	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
25	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
26	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
27	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
28	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
29	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
30	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
40	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
60	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
80	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
100	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
1000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table A.3. Table of the χ^2 distribution for different degrees of freedom df and significance levels α .

df	α level									
	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Figure A.1. Table of the quantiles (p -values) for the Mann-Kendall's S statistic and τ correlation coefficient.
 One-sided: $p = \mathbb{P}[S \geq x] = \mathbb{P}[S \leq -x]$

N = Number of data pairs					N = Number of data pairs				
	4	5	8	9		3	6	7	10
x					x				
0	0.625	0.592	0.548	0.540	1	0.500	0.500	0.500	0.500
2	0.375	0.408	0.452	0.460	3	0.167	0.360	0.386	0.431
4	0.167	0.242	0.360	0.381	5		0.235	0.281	0.364
6	0.042	0.117	0.274	0.306	7		0.136	0.191	0.300
8		0.042	0.199	0.238	9		0.068	0.119	0.242
10		0.0083	0.138	0.179	11		0.028	0.068	0.190
12			0.089	0.130	13		0.0083	0.035	0.146
14			0.054	0.090	15		0.0014	0.015	0.108
16			0.031	0.060	17			0.0054	0.078
18			0.0156	0.038	19			0.0014	0.054
20			0.0071	0.022	21			0.0002	0.036
22			0.0028	0.0124	23				0.023
24			0.0009	0.0063	25				0.0143
26			0.0002	0.0029	27				0.0083
28			<0.0001	0.0012	29				0.0046
30				0.0004	31				0.0023
32				0.0001	33				0.0011
34				<0.0001	35				0.0005
36				<0.0001	37				0.0002
					39				<0.0001
					41				<0.0001
					43				<0.0001
					45				<0.0001